

	EMERGENCY MEDICAL SERVICES AGENCY	POLICY NUMBER	718
	A Division of the Merced County Department of Public Health	Effective Date	11/2025
Authority: Health and Safety Code, Division 2.5, and California Code of Regulations, Title 22, Division 9, Chapter 4, 1797.220		Initial Date:	01/2017
		Review Date:	01/2028

PAIN MANAGEMENT	
ADULT	PEDIATRIC
BLS	
Assess Vitals / Obtain SpO ₂ / Oxygen. Titrate to SpO ₂ of 94% or higher / Secure Airway - Assist Ventilations, if needed. Treat underlying causes of pain: reposition, bandage, splint, traction, cold pack, elevation.	
ALS	
Follow BLS procedures if applicable. Vascular Access – If indicated. Obtain 4-Lead ECG – (Required for narcotic / If indicated for non-narcotic). Capnography – Patients receiving narcotics.	
Non-Narcotic Medications	
Non-narcotic medications may be used with narcotic medications for severe pain.	
Acetaminophen < 50 kg - 15 mg/kg IV/IO Infusion - Max dose 750 mg ≥ 50 kg - 1000 mg IV/IO Infusion - Administer over 20 minutes - Single Dose Only	Acetaminophen - 15 mg/kg IV/IO Infusion - Max dose 750 mg - Administer over 20 minutes - Single Dose Only
Ketorolac (Toradol) - 15 mg IV/IO <u>OR</u> 30 mg IM/IN - Slow IV push over 15 seconds - Single dose only	Ketorolac (Toradol) - 0.5 mg/kg IV/IO/IM/IN - Slow IV push over 15 seconds - Single max dose 15 mg - Single dose only
Narcotic Medications	
Ketamine AND Fentanyl or Morphine may not be administered to the same patient.	
Fentanyl - 25 - 50 mcg IV/IO/IM - Slow IV/IO may repeat every 5 minutes - IM may repeat once in 10 minutes - Total max dose 200 mcg Fentanyl - 1.5 mcg/kg IN - Single max dose 75 mcg - Repeat once after 10 minutes	Fentanyl - 1 mcg/kg IV/IO/IM - Single max dose 25 mcg IV/IO <u>OR</u> 50 mcg IM - Slow IV/IO may repeat every 5 minutes - IM may repeat once in 15 minutes - Total max dose 100 mcg Fentanyl - 1.5 mcg/kg IN - Single max dose 50 mcg - May repeat once after 10 minutes

Signatures on File

Approved By: Tim Williams
EMS Administrator

Ajinder Singh, MD
EMS Medical Director

Morphine - 2 - 5 mg IV/IO OR 5 - 10 mg IM

- Slow IV/IO may repeat every 5 minutes
- IM may repeat once in 10 minutes
- Total max dose 20 mg

Morphine - 0.1 mg/kg IV/IO/IM

- Single max dose 2.5 mg
- IV/IO may repeat every 5 minutes
- IM may repeat once in 15 minutes
- Total max dose 10 mg

Ketamine - 0.3 mg/kg IV/IO Infusion OR 0.5 mg/kg IM/IN

- 30 mg max dose
- Mix in 100 - 250 mL bag of NS or D5W
- Infuse over 5 minutes
- May repeat once after 15 minutes if pain scale remains 5 or higher

No Standing Order

Contraindications

Fentanyl/Morphine:

- SBP is less than length-based assessment tape target for pediatric age
- SBP is less than 90 mmHg for adults
- Respiration less than 12

Ketamine:

- GCS less than 15
- Pregnancy
- Age less than 15 years old
- Known or suspected alcohol or drug intoxication
- Pain Scale less than 5

Acetaminophen:

- Severe Hepatic Impairment
- Liver disease or transplant

Ketorolac (Toradol):

- Age greater than 65 or less than 2 years of age
- History of Renal Disease or Kidney Transplant
- Hypotension
- Active bleeding / G.I. bleeding or Ulcers
- Acute Trauma
- Headaches suspected from Acute Intracranial Hemorrhage
- Pregnancy
- Allergic to NSAIDS
- Asthma
- Coagulation disorder
- Anticoagulation medications
- Current Steroid use
- Acute Coronary Chest Pain

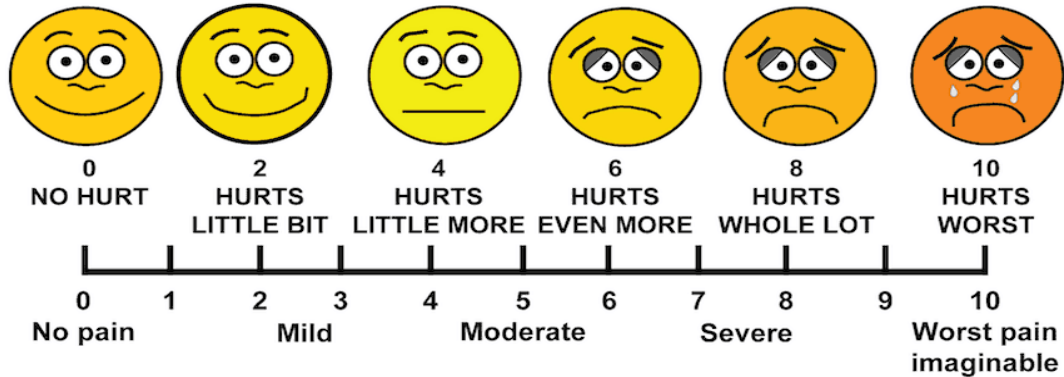
Special Considerations

Administer Fentanyl slowly to prevent Rigid Chest Syndrome. Naloxone may reverse Rigid Chest Syndrome.

Documentation

- Vital signs within 5 minutes pre and post medication administration
- Ketamine Medication Chart must be referenced in documentation if used
- Refer to **EMS Policy #540 – Patient Documentation** for further requirements

PAIN MEASUREMENT SCALE



Merced County EMSA Approved Ketamine Dose Chart

Weight (kg)	IV infusion dose at 0.3 mg/kg	Amount drawn in mL and mixed in a 250 mL NS bag for IV infusion (Ketamine vial concentration is 500 mg/10 mL)	Rounded IV infusion dose	Rounded IM/IN dose (mg)	Rounded IM/IN dose (mL)
40 kg	12.0 mg	0.24 mL	0.25 mL	20.0 mg	0.40 mL
45 kg	13.5 mg	0.27 mL	0.25 mL	22.5 mg	0.45 mL
50 kg	15.0 mg	0.30 mL	0.30 mL	25.0 mg	0.50 mL
55 kg	16.5 mg	0.33 mL	0.35 mL	27.5 mg	0.55 mL
60 kg	18.0 mg	0.36 mL	0.35 mL	30.0 mg	0.60 mL
65 kg	19.5 mg	0.39 mL	0.40 mL	30.0 mg	0.60 mL
70 kg	21.0 mg	0.42 mL	0.40 mL	30.0 mg	0.60 mL
75 kg	22.5 mg	0.45 mL	0.45 mL	30.0 mg	0.60 mL
80 kg	24.0 mg	0.48 mL	0.50 mL	30.0 mg	0.60 mL
85 kg	25.5 mg	0.51 mL	0.50 mL	30.0 mg	0.60 mL
90 kg	27.0 mg	0.54 mL	0.55 mL	30.0 mg	0.60 mL
95 kg	28.5 mg	0.57 mL	0.55 mL	30.0 mg	0.60 mL
100 kg	30.0 mg	0.60 mL	0.60 mL	30.0 mg	0.60 mL
>100 kg	30.0 mg	0.60 mL	0.60 mL	30.0 mg	0.60 mL